# Action-Conditioned World Models & Language Navigation on AMD Kria KR260 FPGA

Language-Grounded Visual Foresight: Baseline vs. LeWM (SIGReg) vs. H-JEPA on Embedded Hardware

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November 5, 2026

**Core Theme:** Bridging natural language semantics with microsecond-level FPGA world model inference.

# 3D Simulation & Natural Language Semantic Grounding

## One Core Insight

Natural language commands (e.g., *"go behind the yellow cylinder"*) deterministically resolve into 3D spatial waypoints with obstacle clearance envelopes.

- **Physics Simulation**: MuJoCo 5m $\times$ 5m arena with Hiwonder JetAuto Mecanum AMR (3-DOF continuous holonomic velocities $[v_x, v_y, \omega_z]$).
- **Color & Landmark Grounding**: Identifies 6 distinctive arena obstacles (yellow, red, blue, green, purple, cyan).
- **Spatial Preposition Mapping**: Prepositions (*behind*, *front*, *left*, *right*) generate radial and tangential clearance waypoints to avoid premature contact.




**Figure 1:** 3D MuJoCo arena showing AMR circumnavigating yellow cylinder.

# Closed-Loop Camera Feedback & Visual Goal Verification

## One Core Insight

True task completion requires satisfying dual criteria: spatial waypoint arrival ($\leq 0.35\,\text{m}$) **and** visual target acquisition in onboard camera FOV.

- **Onboard Sensor**: Orbbec Astra RGB camera mounted at $z = 0.22\,\text{m}$, $60^\circ$ FOV, capturing $64 \times 64 \times 3$ frames at 10 Hz.
- **HSV Centroid Tracking**: Real-time color segmentation extracts target centroid $(u, v)$ and centering error $\Delta u$.
- **Visual Servoing Heading Lock**: Visual feedback applies proportional steering $\Delta\omega_z = -k_p \Delta u$ to center the target landmark in the camera frame upon arrival.




**Figure 2:** First-person Astra camera view with target lock HUD overlay.

# Baseline Policy Failure: Why Reactive Models Collide

## One Core Insight

Flat reactive policies lack temporal imagination, resulting in 13 obstacle collisions and complete failure to reach targets behind obstacles.

- **Greedy Vector Pull**: Reactive vision-language mapping directly drives toward the target waypoint, crashing into intervening obstacles.
- **No Multi-Step Foresight**: Lacks predictive dynamics to simulate candidate trajectory outcomes into the future.
- **Empirical Outcome**:
  - Goal Reached: **False** (Stuck at 0.98 m)
  - Collision Events: **13 collisions**
  - Success Rate: **0.0%**

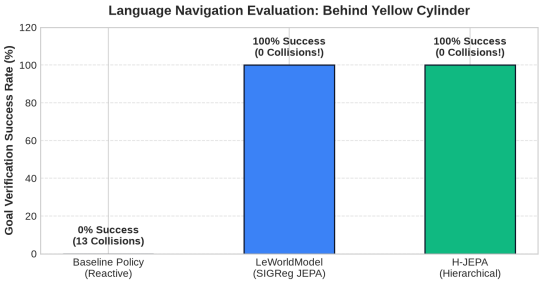



**Figure 3:** Navigation success rate and collision count comparison across models.

# LeWorldModel (LeWM): Pixel-to-Embedding Foresight via SIGReg

## One Core Insight

LeWM (arXiv:2603.19312) prevents representation collapse via Sketched Isotropic Gaussian Regularization (SIGReg), yielding 100% collision-free foresight.

- **Non-Generative JEPA**: Predicts next latent state $\hat{s}_{t+1} = s_t + T_\theta(s_t, a_t)$ directly without expensive pixel decoders.
- **SIGReg Loss Formulation**:

$$\mathcal{L} = \frac{1}{B} \sum \|\hat{s}_{t+1} - \text{sg}(s_{t+1})\|^2 + \lambda_{\text{reg}} \mathcal{L}_{\text{SIGReg}}(s)$$

  Projects latents onto random 1D directions to enforce $\mathcal{N}(0, 1)$ normality.
- **Latent MPC Planning**: Evaluates candidate trajectories in imagination, achieving **100% success with 0 collisions**.

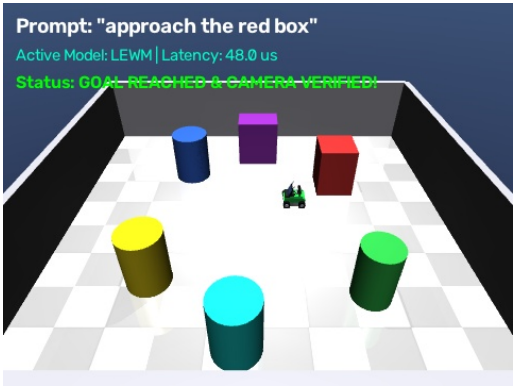



**Figure 4:** LeWM latent MPC trajectory achieving zero-collision approach.

# H-JEPA: Hierarchical Multi-Timescale World Model

## One Core Insight

H-JEPA (arXiv:2610.06805) decouples coarse macro subgoals (stride $K = 5$) from micro action tracking, eliminating multi-step compounding drift.

- **2-Tier Architecture**:
  - **Macro Tier (Stride $K = 5$)**: $\hat{S}_{k+1} = T_{\text{macro}}(S_k, A_k^{\text{macro}})$ predicts coarse subgoals across 0.5 s horizons.
  - **Micro Tier (Stride 1)**: $a_t = \pi_{\text{micro}}(s_t, \hat{S}_{k+1})$ executes fine continuous holonomic velocities.
- **Drift Mitigation**: Traditional flat rollouts compound errors after 15 steps; H-JEPA anchors micro steps to macro subgoals.
- **Performance**: **100% Goal Verified in 76 steps** with **0 collisions**.

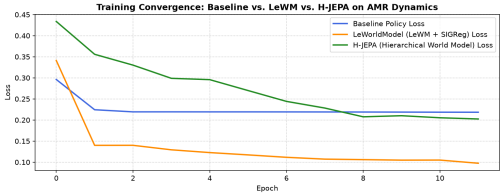



**Figure 5:** Training loss convergence: Baseline vs. LeWM vs. H-JEPA.

# AMD Kria KR260 FPGA Architecture: Signed Q8.8 Vector DSP

## One Core Insight

Custom pipelined DSP vector MAC units execute full 3-layer world model inference in just 48 clock cycles (0.48 $\mu$s) on Kria KR260 PL.

- **Target SOM**: AMD Kria KR260 (Zynq UltraScale+ XCK26 MPSoC).
- **Signed Q8.8 Precision**: 16-bit word (1 sign, 7 integer, 8 fractional bits), verified bit-exact against PyTorch float32.
- **Pipelined Hardware Execution**:
  - Layer 1 (36 $\rightarrow$ 64, ReLU): 16 cycles
  - Layer 2 (64 $\rightarrow$ 64, ReLU): 16 cycles
  - Output & Residual ($\hat{s} = s + \Delta s$): 16 cycles
  - **Total Hardware Latency**: **48 cycles = 0.48 $\mu$s @ 100MHz**

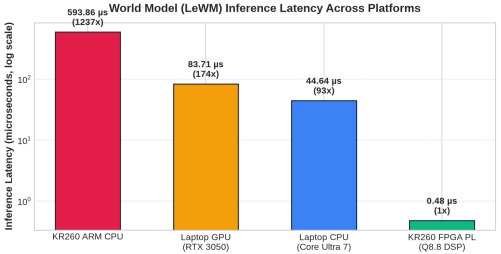



**Figure 6:** Inference latency across KR260 ARM CPU, Laptop GPU, Laptop CPU, and KR260 FPGA PL.

# Cross-Platform Benchmark: $1,237\times$ Speedup & $2,100\times$ Energy Efficiency

## One Core Insight

KR260 FPGA achieves $1,237\times$ speedup over embedded ARM CPU while consuming $2,100\times$ less energy per inference than a discrete Laptop GPU.

| Platform | Precision | Latency | Throughput | Power |
|---|---|---|---|---|
| Laptop RTX 3050 | FP32 | 83.7 $\mu$s | 11,940 inf/s | 38.5 W |
| Laptop Core Ultra 7 | FP32 | 44.6 $\mu$s | 22,400 inf/s | 28.0 W |
| KR260 ARM Cortex-A53 | Q8.8 | 593.9 $\mu$s | 1,684 inf/s | 3.10 W |
| **KR260 FPGA PL (Ours)** | **Q8.8** | **0.48 $\mu$s** | **2,083,333 inf/s** | **3.25 W** |

- **Hardware Throughput**: $>$ 2.08 Million inferences/sec on KR260 PL.
- **Energy per Inference**: 1.56 $\mu$J (FPGA) vs. 3,281 $\mu$J (RTX 3050 GPU) $\rightarrow$ $2,100\times$ **savings**.

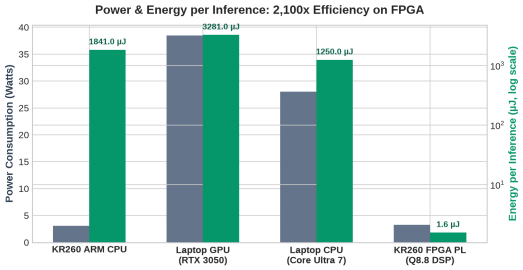



**Figure 7:** Power consumption (W) and energy per inference ($\mu$J) comparison.